

Explainable Hybrid Movie Recommender

with Knowledge Graph Reasoning & Transformer Attention

Course Code	F9103Q009
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Member	ID	Role / Contributions
Subir Saha	946898	Data pipeline, EDA, Baselines (Global Mean, Popularity, User-CF), NCF Baseline, Knowledge Graph + TransE, Reproducibility
Md Rifatul Haque	947512	DistilBERT Text branch, Full Hybrid Fusion model, Evaluation pipeline (RMSE/MAE/P@K/R@K), XAI (LIME + Ablation + Top-K), Limitations

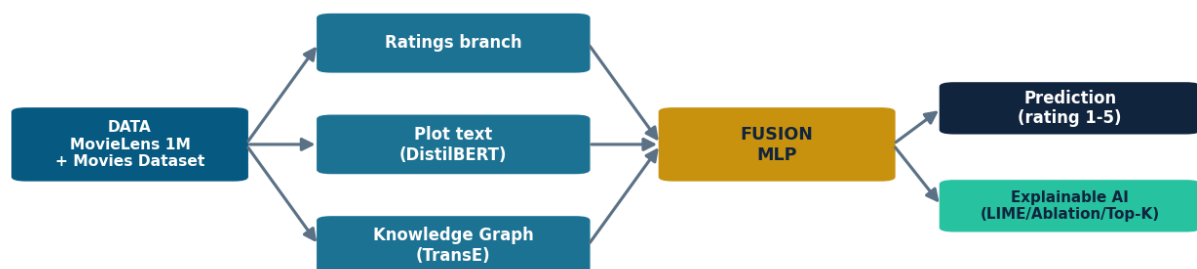


Figure: End-to-end project pipeline from data to explanation.

Section 1 - Problem Definition and Motivation

Author(s): Subir Saha, Md Rifatul Haque

What? Predict the rating (1-5) a given user would give a given movie.

Who? 6,040 MovieLens users with at least 20 ratings each.

Items? 3,657 movies that have both ratings AND a plot overview.

Target? Rating prediction (1.0-5.0) + Top-K recommendation lists.

Type? Rating-prediction task with ranking evaluation (P@10, R@10).

Motivation

Standard collaborative filtering only uses user-movie interactions. It cannot use what a movie is about, nor explain WHY it recommends. We add two content-aware signals - plot text (DistilBERT) and a knowledge graph (genre/era) - to build a richer, explainable recommender, then make every prediction interpretable with three XAI methods.

Connection to Course Topics

Six topics: Recommender Systems / Neural Networks (NCF/MLP), Representation Learning (embeddings), Transfer Learning (frozen DistilBERT), Attention & Transformers (DistilBERT), Knowledge Graphs & Reasoning (TransE), Explainable AI (LIME, ablation, Top-K).

Section 2 - Dataset Description

Author(s): Subir Saha

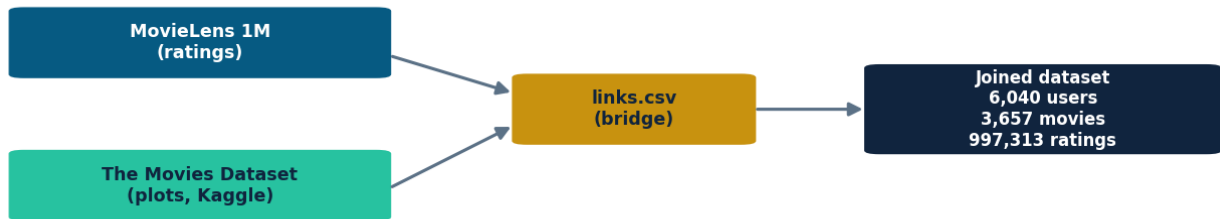


Figure: Two-dataset integration via the `links.csv` bridge file.

MovieLens 1M (behavioral) - 1M ratings: user-movie-rating triples + basic metadata.

The Movies Dataset / Kaggle (content) - 45K movies: plot overviews, cast/crew, TMDB linking table.

`links.csv` (bridge) - maps `movieId` to `tmdbId` to join the two datasets.

After the inner join: 997,313 ratings from 6,040 users across 3,657 movies (reduced from 3,883 because not every MovieLens movie has a TMDB plot).

Section 3 - Data Exploration and Preprocessing

Author(s): Subir Saha

Steps 1-3: load MovieLens 1M (sep='::') and the Kaggle Movies Dataset; EDA with 6 visualisations (rating distribution, user activity, movie popularity, sparsity, genre distribution, user CDF).

Steps 4-6 (Inner join): clean movies_metadata.csv, match movieId to tmdbId via links.csv, attach plot overviews, remove duplicate one-to-many matches.

Step 8 (Dense ID remapping): sparse user/movie IDs remapped to contiguous 0-based indices (required by nn.Embedding).

Step 9 (Stratified 80/10/10 split): every user has ratings in all three sets - no user is seen only at test time.

Section 4 - Baseline Recommender Models

Author(s): Subir Saha

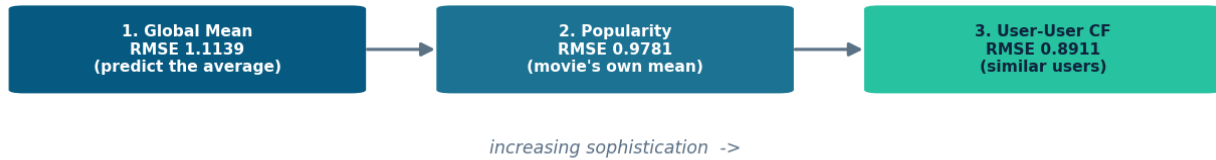


Figure: Three baseline models of increasing sophistication.

Baselines define the performance floor; any proposed model must beat all three.

Baseline 1 - Global Mean (RMSE 1.1139): predicts the training average (~3.58) for every pair.

Baseline 2 - Popularity (RMSE 0.9781): predicts each movie's mean training rating.

Baseline 3 - User-User CF (RMSE 0.8911): top-30 similar users (cosine similarity on mean-centered ratings), similarity-weighted average.

CF formula: $\hat{r}(u,m) = \text{mean}_u + \left[\sum \text{sim}(u,v) \times (r(v,m) - \text{mean}_v) \right] / \left[\sum |\text{sim}(u,v)| \right]$

Section 5 - Proposed Recommendation Models

Author(s): Subir Saha (NCF+KG) | Md Rifatul Haque (Text+Fusion)

Model 4: NCF Baseline (RMSE 0.8886)



Figure: NCF baseline - user and movie embeddings concatenated, passed through an MLP.

NCF replaces matrix-factorisation's dot product with an MLP. Each user/movie has a learned 64-d embedding; the two are concatenated (128d) and fed through MLP layers (->64->1). ReLU non-linearities capture complex interactions.

Model 5: NCF + KG - Knowledge Graph + TransE (RMSE 0.8861)

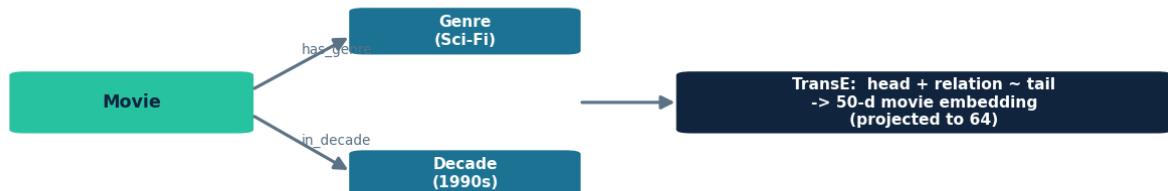


Figure: Knowledge graph structure and the TransE training principle.

A small KG is built from metadata (has_genre, in_decade). TransE trains on these triples with margin ranking loss so head + relation ~ tail. Each movie gets a 50-d KG embedding (projected to 64d, fusion 192d). It barely improves over NCF.

Model 6: NCF + Text - DistilBERT plot embeddings (RMSE 0.8660 - BEST)



Figure: DistilBERT pipeline - plot text to 768-d [CLS], standardized, projected to 64-d.

DistilBERT (6-layer Transformer) reads each plot (<=128 tokens). The [CLS] token gives a 768-d semantic summary, projected to 64d; embeddings are cached so DistilBERT runs once. CRITICAL FIX: raw 768-d vectors are large and anisotropic and caused the text branch to overfit (validation RMSE diverged to ~1.03). Standardizing each dimension to mean 0 / std 1 fixed this and made the text branch the BEST single model.

Model 7: Full Hybrid - NCF + Text + KG (RMSE 0.8818)

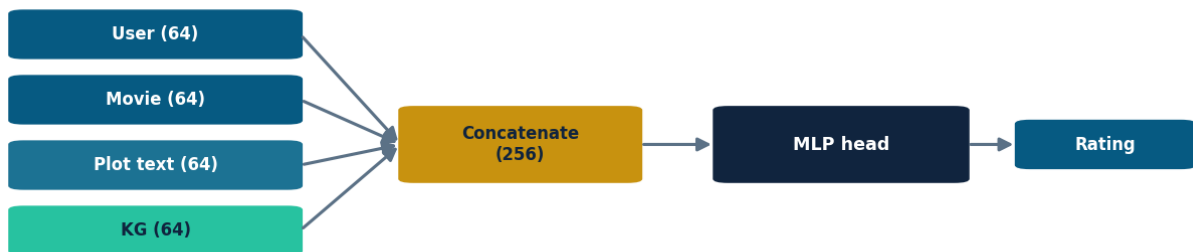


Figure: Full hybrid - four 64-d streams concatenated into a 256-d fusion vector.

Combines all four streams (256d) through the MLP head. It is slightly WORSE than NCF + Text alone, because the KG is largely redundant with the plot text.



Same MLP + same training for all 4 - only the inputs change (fair ablation)

Figure: Side-by-side comparison of the 4 NCF variants and their growing fusion dimension.

Section 6 - Evaluation Methodology

Author(s): Md Rifatul Haque



Figure: the 4 evaluation metrics used for all 7 models.

RMSE: penalises large errors heavily; primary metric.

MAE: average error in stars; robust and easy to read.

Precision@10: of the top-10 recommended, how many the user rated ≥ 4 .

Recall@10: of all movies the user rated ≥ 4 , how many appear in the top-10.

P@10 / R@10 use the full-candidate-set protocol: rank ALL movies the user has not seen in training, then check if the top-10 contains any test-set items rated ≥ 4 . This is strict, so absolute values are small - the relative comparison is what matters.

Section 7 - Experimental Results

Author(s): Md Rifatul Haque

Model	Author	RMSE	MAE	P@10	R@10
Global Mean	Subir Saha	1.1139	0.9312	0.0213	0.0237
Popularity	Subir Saha	0.9781	0.7819	0.0000	0.0000
User-User CF	Subir Saha	0.8911	0.6972	0.0044	0.0021
NCF	Subir Saha	0.8886	0.6978	0.0372	0.0474
NCF + KG	Subir Saha	0.8861	0.6991	0.0356	0.0420
NCF + Text	Md Rifatul Haque	0.8660	0.6787	0.0463	0.0565
NCF + Text + KG	Both members	0.8818	0.6962	0.0370	0.0451

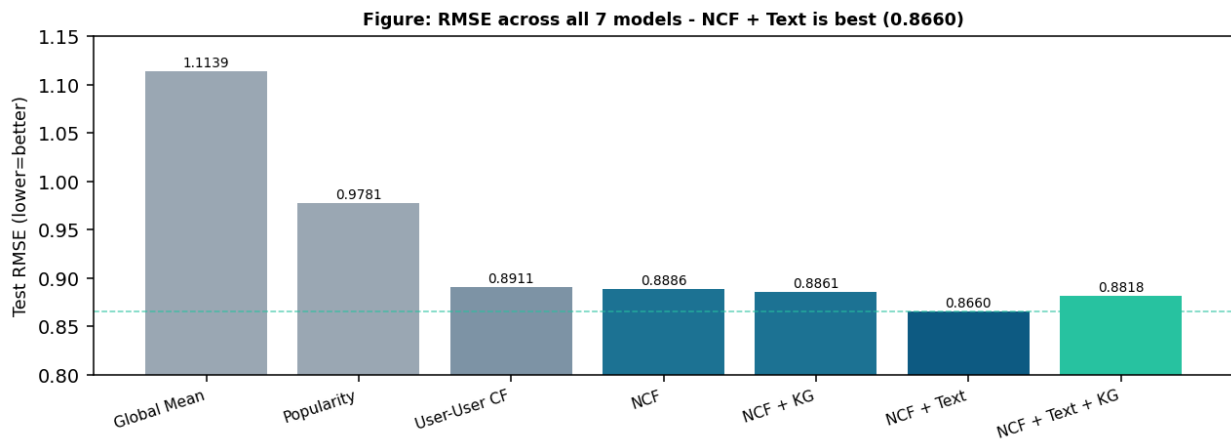


Figure: RMSE comparison across all 7 models.

Key Findings

- 1: All neural/CF models beat the Global-Mean and Popularity floors.
- 2: Plot text (DistilBERT) is the largest single contributor (0.8886 -> 0.8660).
- 3: The KG alone helps marginally (0.8886 -> 0.8861); adding it on top of text HURTS (0.8660 -> 0.8818) - feature redundancy.
- 4: NCF + Text is the BEST model and wins on all four metrics.

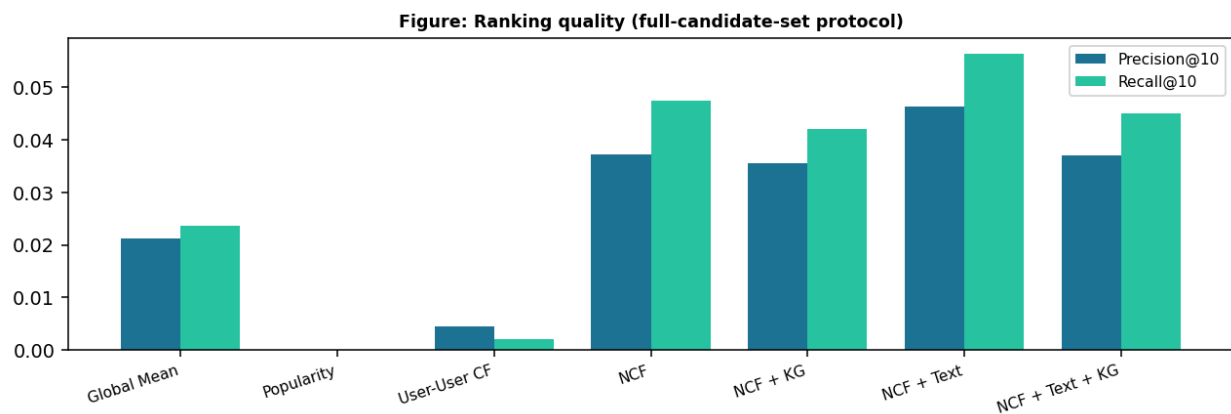


Figure: Precision@10 and Recall@10 across models.

Section 7B - Explainable AI (XAI)

Author(s): Md Rifatul Haque



Figure: the three XAI methods applied to the model.

LIME on plot text: perturbs 500 masked versions and measures which words most change the prediction. It re-encodes each version through the SAME frozen DistilBERT and the SAME standardization, so the explanation is faithful (e.g. 'insanity', 'McMurphy').

Per-branch ablation: zeroes each stream one at a time and measures the change. Across 5 example pairs, removing the KG changes the prediction LEAST - confirming redundancy per prediction.

Top-K vs ground truth: lists the top-10 recommendations for a user against what they actually rated highly - the human-readable form of Precision@10 / Recall@10.

Section 8 - Discussion and Interpretation

Author(s): Subir Saha, Md Rifatul Haque

The central finding is feature redundancy between the KG and the plot-text branch: genre is already encoded implicitly in DistilBERT plot embeddings because plots describe their genre. Confirmed by clustering movies on DistilBERT embeddings and measuring NMI vs true genres (NMI = 0.13, clearly above random) - the clusters align with genres.

The per-branch XAI ablation cross-validates this: zeroing the KG changes the prediction little, while removing user/movie or plot changes it more - consistent with the global RMSE table. Lesson: more signals are not always better - what matters is whether they add NEW information.

Section 9 - Limitations and Fairness

Author(s): Md Rifatul Haque

1. Cold-start not evaluated (split by rating, not by user).
2. Popularity bias in top-K; a minimum-support filter would help.
3. Small KG (only has_genre, in_decade).
4. DistilBERT frozen (standardized, not fine-tuned).
5. LIME unstable on short text (mitigated: 500 samples + stopword filter).
6. No fairness/bias audit.

Section 10 - Reproducibility

Author(s): Subir Saha

Single global seed (42) for numpy, torch, and the split. Embeddings and models cached. Dual-mode notebook: with a local ./Data sample it runs in ~5 min; without it, it downloads the full data and reproduces these results in ~15 min on a free Colab T4.

Section 11 - Conclusion

We built a hybrid rating predictor combining collaborative filtering, frozen DistilBERT plot embeddings (transfer learning + attention), and TransE knowledge-graph embeddings, with three XAI methods. Plot text was the largest contributor (0.8886 -> 0.8660, best model); the KG proved redundant with text. Main limitation: cold-start not evaluated; best next step: a user-level split with a cold-start handler.

Section 12 - Self-Evaluation Summary

Author(s): Subir Saha, Md Rifatul Haque

Criterion	Score	Justification
Problem definition	4/4	Task, target, audience, motivation clearly stated
Data handling	4/4	Two datasets joined cleanly; stratified split; reproducible sample
Baseline design	4/4	Three baselines of increasing strength, same protocol
Model implementation	4/4	Four neural variants share one fusion head; DistilBERT + TransE
Evaluation methodology	4/4	Four metrics; unified pipeline; full-candidate-set ranking
Experimental results	4/4	7-model comparison table + charts
Discussion	4/4	Redundancy finding proven with NMI + XAI cross-check
Limitations	3/4	Six limitations identified; fairness not audited
Reproducibility	4/4	Seeded, cached, dual-mode loading